

Governance Structure of the U24 WPP Research and Coordination Center

- 1) **Bidirectional Communication:** The governance of the NCCIH Cooperative Agreements will involve continuous, bidirectional communication between the funded Whole Person Reference Physiome Research and Coordination Center (WPP-RCC) and the NCCIH and NIH staff (refer to Figure 1).
- 2) **Progress Reporting:** The progress of the WPP Research and Coordination Center will be reported to senior NCCIH leadership, including the NCCIH Director, NCCIH Division Director, and NCCIH Branch Chief, through the Cooperative Agreement's Program Officer (PO).
- 3) **Steering Committee Composition and Responsibilities:** The WPP-RCC Steering Committee is composed of the WPP-RCC PI Leadership Team and the WPP-RCC NIH Program Team. The Steering Committee will oversee the formation and evaluation of five WPP-RCC Working Groups, ensuring the successful development of the WPP-RCC Resource Center. Each Steering Committee member has one vote in decision-making processes.
- 4) **Steering Committee Chair:** The Chair of the Steering Committee should be the Contact PI of the WPP-RCC U24 award. The Chair is responsible for organizing monthly Steering Committee meetings and developing the meeting agendas. Additional meetings may be requested by the PO or SC Chair if necessary.
- 5) **Voting Process:** Official Steering Committee members will vote on critical issues related to U24 operations. NIH votes shall not exceed one-third (1/3) of the total votes in the Steering Committee when decisions are made through voting.
- 6) **PI Leadership Team:** The WPP-RCC PI Leadership Team is composed of the PI and Multi-PIs who will serve as official members of the Steering Committee, including Drs. Snyder, Pei, Susztak, Börner (MPIs), and Weber (consultant).
- 7) **NIH Program Team:** The WPP-RCC NIH Program Team will serve as Subject Matter Experts (SMEs) or Project Scientists (PS) and is composed of trans-NIH POs. NCCIH will assign a Program Officer and at least one PS or SME to the U24 Whole Person Research and Coordination Center. Each participating NIH Institute, Center, or Office (ICO) may designate scientific or program staff as PSs or SMEs to the U24. Program staff from participating ICOs may also serve as NIH representatives on core working groups or as Project Scientists for the NIH Whole Person Initiative. Additionally, the NLM will provide technical expertise to ensure compatibility of the Whole Person Physiome Conceptual Map with HuBMAP and the Human Reference Atlas.
- 8) **Working Groups:** There are four WPP-RCC Working Groups (WGs): Scientific Leadership WG, Expert Domain Mapping and Common Data Element WG, Data Science and Modeling WG, and Coordination WG. Each WG, except for the Coordination WG, will have its own Consultation Committee composed of members external to the WG and possibly outside the initial grant application personnel.
- 9) **External Advisory Board (EAB):** EAB members will be nominated by the Whole Person Leadership Team and/or NCCIH leadership, but the EAB members list needs to be reviewed and approved by NCCIH Leadership. The EAB will report to and advise NCCIH leadership. Whole Person Physiome Research and Coordination Center (WPP-RCC) governance structure is outlined in (**Figure 1**).

Core 1 – Scientific Leadership Core is divided into three parts: a) Leadership Team (composed of the leaders of each of the four Cores: Drs. Snyder, Pei, Susztak, Börner [MPIs], Weber [consultant]), b) Advisory Board, to be composed of 5 experts in modeling, physiology, ontology, and ethics approved by the NIH, c) NIH stakeholders. Core 1 will oversee the direction of the WPP-RCC, organize the WPP-RCC team, and oversee the development of the WPP-RCC portal. A dedicated Data Manager in the Snyder Lab, will construct the portal.

Core 2 – Expert Domain Mapping and Common Data Element Core will produce the WPP conceptual map (Objective 1). Core 2 will be organized into **Core 2a – Expert Domain Mapping** led by Drs. Pei and Susztak to provide the expert conceptual map and **Core 2b – Common Data Element** led by Drs. Börner and Weber to organize the conceptual map into a knowledge graph. Core 2 will be advised by the Whole Person Working Group [layer one domain expert coordinators: 11 organ-expert Co-Is + MPI Dr. Susztak; layer two broader domain experts].

Core 3 - Data Science and Modeling Core will produce the WPP prototype model (objective 2 of the RFA) and will be led by Dr Snyder. Core 3 will collaborate with the broader modeling community.

Core 4 – Coordination Core will manage collaboration between the WPP-RCC team and outside scientists, and support Core 1 in facilitating WPP team collaboration. Core 4 will be led by Dr. Snyder and supported by a program manager in the Snyder Lab.

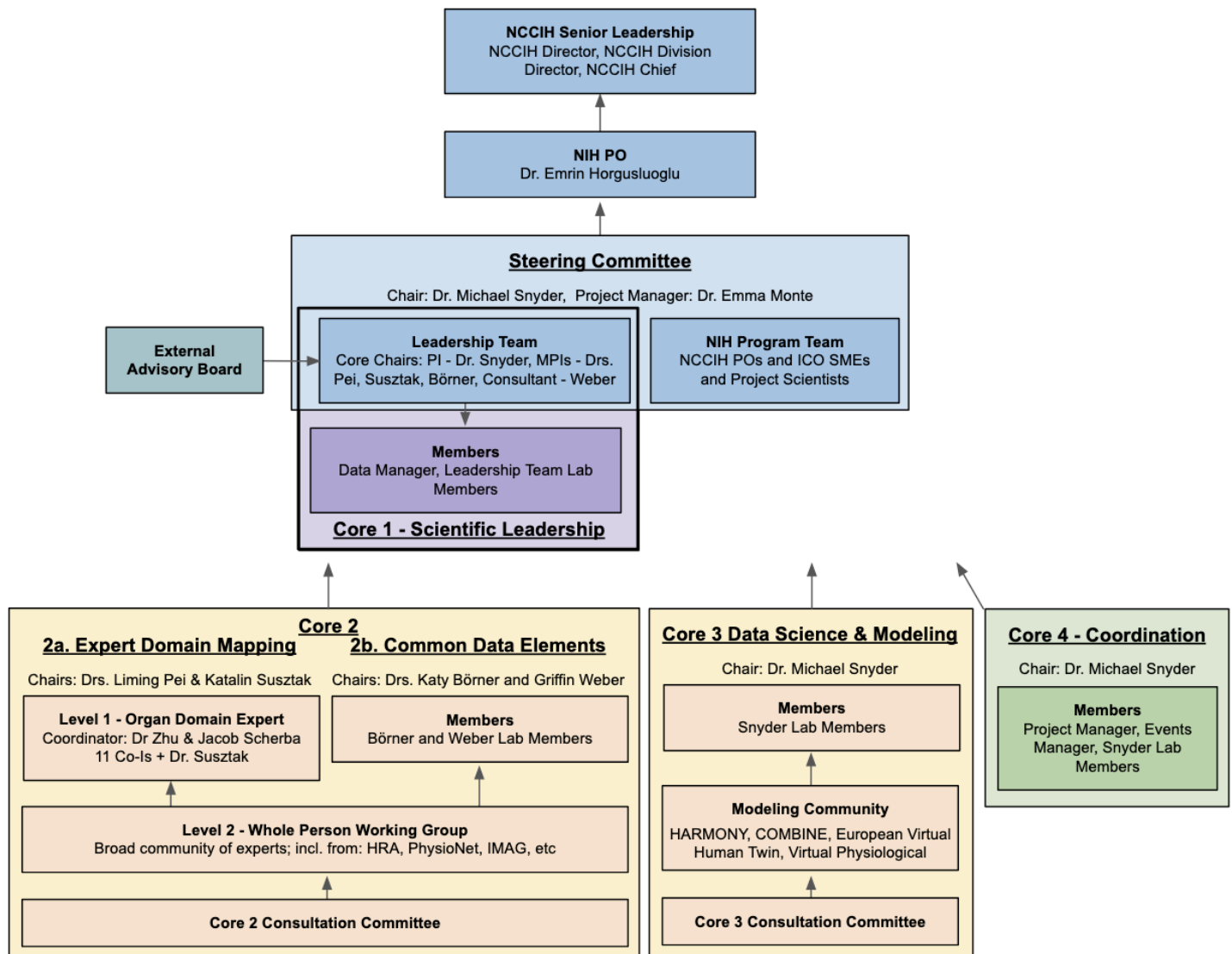


Figure 1. Organization of the Whole Person Physiome Research and Coordination Center.

Decisions of the four Cores will be made by consensus of the Core Leaders. In the unlikely event of differences of opinion, the entire Core 1 Leadership Team will weigh in on a decision of the Core. Decisions affecting the entire WPP-RCC will be made via discussion and consensus amongst the Core 1 Leadership Team

with input from the Advisory Board, NIH stakeholders, and, where relevant, Whole Person Working Group. If necessary, the Advisory Board will serve as the tie-breaker.

Expected Regular Meetings

- **Leadership Team:** Weekly virtual meetings (Chairs of the four Cores + lab members).
- **Steering Committee:** Monthly virtual meetings (Leadership Team + NIH Stakeholders).
- **Core 2 Leadership:** Weekly virtual meetings (Chairs of Core 2a: Expert Domain WG and Core 2b: CDE WG).
- **Organ Domain Expert:** Regular in-person meetings (Core 2a Chairs + Level-1 organ leads).
- **Whole Person Working Group:** Monthly virtual meetings (Level 1 and Level 2 domain experts
Level 1 experts are a small team of one MPI and 11 Co-Is with 1 representative for each of the major organ systems. These are MD clinician scientists whose job is to guarantee we are recruiting the best experts for each organ and coordinate the map building for their organs. Level 2 experts are a large group of experts with expertise in either Core 2a or Core 2b domains recruited to collaborate together on the map. They are recruited by Level 1 experts and the Core 2 leadership team to provide additional domain expertise on the physiology of each organ, cross-organ physiology, common data elements, and knowledge graphs.).
- **Whole Person Working Group Workshops:** Monthly virtual meetings (Level 1 and Level 2 domain experts).
- **Core 3 Team:** Weekly virtual meetings.
- **Whole Person Annual Meeting:** Annual in-person meeting (all of WPP-RCC and outside scientists).
- **Whole Person Data Jamboree:** Annual virtual meeting (WPP and outside scientists).
- **Additional Working Group Meetings:** TBD (Subprojects emerging from Annual Meeting may form working groups to continue to collaborate through regular virtual meetings).
- **Additional Collaborative Meetings:** Proposed additional meetings with outside groups, including regular conferences in the modeling community as highlighted in our Milestones.
- **Advisory Board:** Invited to participate in Steering Committee Meetings and the Annual Meeting.